Integral images of Springer–Borel maps

Connection with Parts II and III

Discussion note prepared for Shrawan Kumar

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Purpose and status. This note summarizes an integral refinement of the maps studied in Kumar–Rogers, Part II [KR20], and Kumar–Xie, Part III [KX24]. It separates published statements, deductions whose arguments are recorded and checked in the companion repository, and unproved conjectures. Deductions are presented with their arguments in the companion notes; external expert review and originality remain unestablished.

1 The published starting point

Published result/input. Part II gives the classical symplectic and odd-orthogonal generator images and stable injectivity: Theorem 16 for C and Theorem 29 for B. Part III gives the even-orthogonal generator images, its stable integral presentation, and stable injectivity, especially Theorems 3.6 and 3.8 and Proposition 3.7. The injectivity theorems are stated over $\mathbb{C}$; their proofs explicitly use chosen integral forms.

Use the normalized torus coordinates $\bar{t}_j = (t_j - t_j^{-1})/2$, corresponding to x_j under the Springer–Borel map. Put $c_i = c_i(Q)$, where Q is the tautological quotient bundle on the relevant isotropic Grassmannian, with $c_0 = 1$ and $c_i = 0$ outside its rank. The squared-tail generator image is

$$e_i(\bar{t}_{\mathrm{tail}}^2) \longmapsto c_i^2 + 2\sum_{j=1}^{i}(-1)^j c_{i+j}c_{i-j}. \tag{1}$$

The integral theta/eta presentations and B/C Schubert comparison used below are existing Schubert calculus [BKT-IG, BKT-D].

2 The additional integral question

Write $n = a + k$. On $X_B = \mathrm{OG}(a, 2n+1)$ use $q_i = e_i(x_{a+1}^2, \ldots, x_n^2)$, $1 \le i \le k$. On $X_D = \mathrm{OG}(a, 2n)$ use $q_1, \ldots, q_{k-1}$ and $p = x_{a+1}\cdots x_n$, with $p^2 = q_k$. Take $a, k \ge 1$ in B and $a \ge 1, k \ge 2$ in D.

Polynomial degree d means cohomological degree $2d$; $\deg q_i = 2i$ and $\deg p = k$. Let $L_d \subset H^{2d}(X;\mathbb{Z})$ be the integer span of all degree-d monomials in these fixed inputs after applying the Borel map. The object to determine is

$$\mathrm{sat}(L_d)/L_d, \qquad \mathrm{sat}(L_d) = (L_d \otimes \mathbb{Q}) \cap H^{2d}(X;\mathbb{Z}). \tag{2}$$

We retain every nonzero Smith factor of $L_d \hookrightarrow \mathrm{sat}(L_d)$, including 1: this means the finite group (2) together with $\operatorname{rank} L_d$. The index is in the selected rational image, whose rank can be smaller than ambient cohomology. Injectivity alone does not imply saturation.

3 Structural deductions and finite type-B formulas

Deduction in the companion notes. The arguments recorded in [Repo] yield the following refinements of the image problem, in the normalization of Section 2. They use published integral presentations; these exact Smith conclusions are not attributed to those sources as published theorems.

Statement	Type B	Type D
Possible nonzero factors	$2^e,\ 0 \le e \le \min(a,k)$	Powers of 2
Stable image, k fixed	Saturated degreewise	Saturated degreewise
Range with all factors 1	$d \le \max(a,k)$	$d \le a$
Complete closed families	$\min(a,k) \le 3$	$a = 1$ (quadrics)

Stable saturation strengthens the injectivity established in the papers: the normalized image contains every integral class in its rational span. The finite ranges are sufficient, not claimed optimal. In B, the full Smith multiset is also unchanged under $a \leftrightarrow k$.

Why the B comparison controls the prime two. For the same squared-root Schur inputs, let C be the integral symplectic coefficient matrix. Its selected image is primitive by the quaternionic Grassmannian fibration and integral Schur basis [PR]. For signed Schubert indices w,

$$B_{w,\lambda} = 2^{\nu(w)} C_{w,\lambda}, \qquad \nu(w) = \#\{\text{negative signed entries}\}. \tag{3}$$

After block duality, $\nu \le \ell := \min(a,k)$. As the full C row lattice is primitive, the B row lattice lies between $2^\ell \mathbb{Z}^R$ and $\mathbb{Z}^R$, where R is the source rank in the degree under consideration. This gives the B exponent bound. The stable B/D saturation arguments instead use integral specialization of the theta/eta models. In D, working over $\mathbb{Z}[1/2]$ with a sign-averaging retraction rules out odd-prime factors. For later use, in the smaller-head presentation define the prefix lattice $P_t = \operatorname{span}_{\mathbb{Z}}\{C_{w,\bullet} : \nu(w) \le t\} \subseteq \mathbb{Z}^R$, with $P_{-1} = 0$. If every prefix is primitive, the multiplicity of 2^t is $\operatorname{rank} P_t - \operatorname{rank} P_{t-1}$.

A concrete closed type-B family. Set $A = \max(a,k)$. For $\ell = 2$, put $u = \lceil A/2 \rceil$, $v = \lfloor A/2 \rfloor$. In polynomial degree $2m$, the complete multiset is

$$\left\{ 2^{e(r,s)} : 0 \le s \le r \le A,\ r+s = m \right\}, \qquad e(r,s) = \begin{cases} 0, & r \le u,\ s \le v, \\ 2, & s > u, \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

The proof uses two-variable plethysm, theta Giambelli, and an integral change of basis with disjoint Schubert supports. For $\ell = 3$, all four multiplicity polynomials likewise have closed formulas, derived through primitive Schur-product prefixes [Repo]. For example, $a = k = 2$ in polynomial degree 4 has factors $(1,2)$: the image has index 2 in its rank-two saturation.

4 Unit factors, classical counting, and the largest factor

For type B put

$$\ell = \min(a, k), \quad A = \max(a, k), \quad r = \lfloor n/2 \rfloor, \quad c = \lceil n/2 \rceil.$$

Let $M_e(m)$ count factors 2^e in polynomial degree $2m$, equivalently cohomological degree $4m$. The variable q below counts m; it is not a generator q_i. The Gaussian $\begin{bmatrix} v \\ u \end{bmatrix}_q$ counts partitions in the rectangle $(v-u)^u$, and is zero outside $0 \le u \le v$. Set $U_{0,0} = 1$.

The polynomial is already known; its Smith interpretation is additional

Published result/input. The polynomial

$$U_{n,\ell}(q) = \sum_{j=0}^{\lfloor \ell/2 \rfloor} q^{j(c-\ell+j)} \begin{bmatrix} r \\ j \end{bmatrix}_q \begin{bmatrix} c-j \\ \ell-2j \end{bmatrix}_q \tag{5}$$

is classical Birkhoff/Hall-type enumeration [B35, Sage]. For ambient type $(2^r, 1^{n \bmod 2})$ and subgroup type $(2^j, 1^{\ell-2j})$, the standard subgroup formula gives its summand. The same expression counts invariant subspaces of a maximal-rank square-zero operator. It is not a new counting polynomial.

Deduction in the companion notes. The integral comparison in the companion notes identifies

$$\sum_m M_0(m) q^m = U_{n,\ell}(q). \tag{6}$$

This is the additional Smith assertion, rather than Birkhoff's theorem. A brief account of the argument is:

1. Modulo 2, only the $\nu = 0$ rows of (3) remain. Littlewood's two-quotient identity identifies their row lattice with a bounded Littlewood–Richardson product lattice.
2. Let $W = \{U \in \mathrm{Gr}(\ell, n) : NU \subset U\}$, with $N^2 = 0$ of rank r. A bounded affine-lattice model realizes W as a closed union of Iwahori Schubert cells; the rectangular case is already established in [HNSS].
3. Anderson's integral Chern-generation theorem [A24] makes $H^*(\mathrm{Gr}(\ell, n); \mathbb{Z}) \to H^*(W; \mathbb{Z})$ surjective. A Grassmann-bundle resolution, with degree-one sections over cells, has primitive cohomology pullback. Its Schur coproduct is the LR matrix above. Thus that row lattice is primitive, and its rank is (5).

The complete integral argument, including the odd-n lattice bounds, is supplied in [Repo]; this paragraph is a proof sketch.

A separate all-rank endpoint deduction

An incidence resolution and integral Pieri straightening identify the prefix $P_{\ell-1}$ with the coordinates indexed by $\lambda_\ell \le c-1$. The complementary partitions have every part at least c, giving

$$\boxed{\sum_m M_\ell(m) q^m = q^{\ell c} \begin{bmatrix} r \\ \ell \end{bmatrix}_q .} \tag{7}$$

This argument does not require primitivity of every intermediate prefix.

5 Relation to the Euler generator in Part III

Deduction in the companion notes. The Euler sector on $\mathrm{OG}(a, 2n)$, generated by $pf(q)$, has the same nonzero Smith factors as the normalized B image on $\mathrm{OG}(a-1, 2n-1)$, shifted by polynomial degree k for $a \geq 2$. The proof uses the explicit difference basis in BKT Proposition 2 [BKT-D]; its rational Chern-linearity upgrades integrally because the basis correspondence and the multiplication maps are integral.

Combining the square and Euler sectors requires further integral data. A swapped pair of typed Schubert classes has sum and difference of index 2. The companion two-flag lattice lemma describes this gluing: a paired sector-exponent pair (i, j) contributes exponents

$$(\min(i,j),\ \max(i,j)+1), \tag{8}$$

with the pairing determined by relative flags over $\mathbb{F}_2$, not by the sector Smith lists alone. This is a lattice reconstruction, not a full type-D partition classification. The observed equality of the general D square sector with the B image remains conjectural here, as does a complete type-D partition formula.

Normalization matters. These statements concern the chosen normalized integral forms appearing in the proofs of Parts II and III. They need not equal the literal intersection with the integral representation ring. For the defining/vector SO/Sp character forms discussed in Part I [K16], a degree-d image is 2^d times its normalized counterpart. Consequently each nonzero Smith factor is multiplied by 2^d. No such conclusion is asserted here for a Spin character lattice.

6 The remaining type-B conjecture

Conjecture. Outside the justified cases, the proposed complete formula is

$$\boxed{\sum_m M_e(m) q^m \stackrel{?}{=} q^{ec} \begin{bmatrix} r \\ e \end{bmatrix}_q U_{n-2e,\ell-e}(q), \qquad 0 \leq e \leq \ell.} \tag{9}$$

Recompute the parameters of U at its displayed arguments. This includes (6) and (7), agrees with all multiplicities for $\ell \leq 3$, and has one further justified layer at $\ell = 4, e = 3$.

For a maximal-rank square-zero operator, define $\delta(U) = \mathrm{rank}(N : U \to V/U)$. The right side of (9) is the compactly supported even Poincare polynomial of $\delta(U) = e$, by an elementary stratification in the companion notes. Over finite fields this is the profile (ℓ, e) counting problem already covered by Ram's general theorem [R25]. The unresolved step is its identification with the intermediate *integral Smith* layers, including the required prefix-rank and primitivity comparisons. Counting the strata alone does not prove (9).

Evidence and limits. The formula matches 174 recorded exact Smith lists and 430 prefix comparisons. A separate verifier checks 516 exact integer row-module equalities. These finite checks support the conjecture but do not prove it in arbitrary rank. No originality claim is made for any deduction. The most useful points for further review are the integral comparison behind (6) and the missing intermediate-layer comparison in (9).

References and companion material

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[Repo] *Sage-programs: orthogonal Smith notes and reproducible calculations*. Proof/source audit dated 19 October 2026, baseline commit `0adbc65`; unpublished companion material. Claim-by-claim status and source map. The status map links the structural, unit-factor, largest-factor, filtration, and type-D proof notes and their exact verification reports.